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# The effects of post-stroke upper-limb training with an electromyography (EMG)-driven hand robot

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ABSTRACT

Loss of hand function and finger dexterity are main disabilities in the upper limb after stroke. An electromyography (EMG)-driven hand robot had been developed for post-stroke rehabilitation training. The effectiveness of the hand robot assisted whole upper limb training was investigated on persons with chronic stroke ($n = 10$) in this work. All subjects attended a 20-session training (3–5 times/week) by using the hand robot to practice object grasp/release and arm transportation tasks. Significant motor improvements were observed in the Fugl-Meyer hand/wrist and shoulder/elbow scores ($p < 0.05$), and also in the Action Research Arm Test and Wolf Motor Function Test ($p < 0.05$). Significant reduction in spasticity of the fingers as was measured by the Modified Ashworth Score ($p < 0.05$). The training improved the muscle co-ordination between the antagonist muscle pair (flexor digitorum (FD) and extensor digitorum (ED)), associated with a significant reduction in the ED EMG level ($p < 0.05$) and a significant decrease of ED and FD co-contraction during the training ($p < 0.05$); the excessive muscle activities in the biceps brachii were also reduced significantly after the training ($p < 0.05$).

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## 1. Introduction

The motor functions of the upper limb, including the hand, are highly related to the daily activities, such as feeding, and cleaning. Approximately 70% to 80% of people after stroke have upper-extremity impairments (Nakayama et al., 1994), which greatly affect the independency in their life. However, in rehabilitation of the upper limb, many stroke survivors experienced reasonable motor recovery of their proximal upper limb (e.g., the shoulder and the elbow) but limited recovery at the distal (e.g., the wrist and the fingers) (Chae et al., 2000; Chae and Yu, 1999). As pointed out by Volpe et al. disability from upper limb impairment depends primarily on the loss of hand function and finger dexterity (Volpe et al., 2008). This might be due to the difficulty of managing the movements of the proximal joints and the finger movements at the same time, and the therapist and the patient usually start a training program with the large and proximal joints first. It is possible that incorporating well controlled finger movements during the upper limb training may improve the training effects, especially at the hand.

Impairment after stroke is usually accompanied by abnormalities of spasticity, muscle weakness, and disturbances in muscular coordination mainly reflected by varied muscle co-activation patterns (Dewald et al., 2001). In many persons after stroke, when the physical signs (spasticity, weakness) have been treated effectively, motor impairment associated with abnormal muscle co-activation patterns could still be present and severe (Landau, 1980). Dewald et al. suggested that the primary source of motor dysfunction or global disability in many stroke survivors was abnormal movement coordination, i.e., abnormal muscle co-activation pattern (Dewald et al., 2001). With the quantitative analyses on the electromyography (EMG) recorded from the paretic upper limb in persons after stroke, reduced muscle co-activation patterns were found, and a relatively high correlation of motion at adjacent joints were also observed during isometric contractions in our previous work (Hu et al., 2006) and in the work by (Dewald et al., 1995). The related clinical observation is that the muscle coordination in persons after stroke is stereotyped and not permitting different combinations of muscle activities (Brunnstrom, 1992). For example, the hand movements are always compensated with the excessive wrist, the elbow and even shoulder motions in most of persons after stroke.

Rehabilitation robots not only can provide therapy for long time period, in a consistent and precise manner, but also can facilitate some fine control on the limb movements by motors, such as for

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the fingers. However, most of current rehabilitation robots are designed for large joints, e.g., the elbow and the shoulder. Studies with robots such as the MIT-Manus, the ARM Guide or the MIME demonstrated improved proximal arm function after post-stroke rehabilitation training (Hesse et al., 2003; Krebs et al., 1998; Lum et al., 2006, 2002), whereas these improvements did not show on the distal arm function. Some robots for hand also were developed for post-stroke training, e.g., the Hand Wrist Assistive Rehabilitation Device (HWARD) system (Takahashi et al., 2008), which can provide the hand grasping/opening and wrist extension/flexion movement, and the HapticKnob system for hand gasping and forearm pronation/supination (Lambercy et al., 2011). Rehabilitation assisted with these hand/wrist robots also demonstrated motor improvements in hand functions after the training (Takahashi et al., 2008; Lambercy et al., 2011). However, when using these hand/wrist systems, subjects mainly focus on the hand/wrist practices, but not the whole upper limb movements, e.g., the shoulder and the elbow, which could be one of the reasons that limits the further improvement of the proximal joint functions as reported (Takahashi et al., 2008; Lambercy et al., 2011). Furthermore, most of the evaluations on training effectiveness by robot-assisted rehabilitation were only conducted by the traditional clinical assessments before and after the training (Volpe et al., 2008, 2004). More detailed and quantitative descriptions on the training progress are needed for a better understanding on the treatment-related motor recovery.

Restoration of motor function is task-relevant after stroke (Rensink et al., 2009; Schweighofer et al., 2012; Jiang et al., 2013). Reorganization of surviving neural networks in the brain rather than repair of the damaged tissue is the key for motor functional restoration after stroke, which depends on repeated and intensive physical practice of the paretic limb on the motor skills to be restored (Lin et al., 2012, 2009; Miyai et al., 2006; Murata et al., 2006; Dijkhuizen et al., 2012; Takeuchi and Izumi, 2013). Therefore, physical training related to daily tasks has been widely adopted in post-stroke rehabilitation (Tan et al., 2012). Arm reaching practices associated with hand grasp and release movements are important for daily tasks. In our previous work, an exoskeleton hand robot was developed (Tong et al., 2010; Ho et al., 2011). The hand robot can be easily worn onto the paretic hand of a person after stroke. The system can help the paretic hand perform hand grasp and release according to a subject's voluntary motor intention detected by EMG signal from the residual muscles of the affected side, meanwhile the subject also can conduct the whole upper limb movement at the same time, such as reaching. Voluntary physical experience is a major modulator for the neural plasticity in the nervous system after stroke (Jiang et al., 2013; Lin et al., 2012, 2009; Miyai et al., 2006; Takeuchi and Izumi, 2013; Ding et al., 2011). Involvement of voluntary efforts during rehabilitation training has been found to be more effective than those only with passive motions (Volpe et al., 2004). We proposed the hypothesis that the task-oriented upper limb training with coordinated hand movements assisted by the voluntary EMG-driven hand robot would improve not only the motor function of the distal joints (e.g. the fingers), but also would improve the muscular coordination of the upper limb. We would investigate the training effectiveness on persons with chronic stroke, and monitor the muscular coordination by quantitative analysis on EMG signals across the whole training sessions. Previously, we used EMG to monitor the muscular coordination patterns between different muscle pairs during robot-assisted elbow and wrist training after stroke (Hu et al., 2007, 2009). It was found that the improvement in the coordination could be captured by EMG patterns among the interested muscles through session-by-session investigation. In this work, the rehabilitation progress during the robot hand assisted training also would be monitored by EMG across the training sessions for a quantitative evaluation on the training effects.

## 2. Methodology

### 2.1. The EMG-driven hand robot

Fig. 1 shows the experimental setup of the study. The hand robot have five individual fingers, and each of them is actuated by a linear actuator (Firgelli L12, Firgelli Technologies Inc.) and provides 2 degrees of freedom for each finger (Fig. 1a). The proximal section will rotate around the virtual centre located at the metacarpophalangeal (MCP) joint, whereas the distal section will rotate around the virtual centre located at the proximal interphalangeal (PIP) joint as marked in Fig. 1a for the index, the middle, the ring and the little fingers. The position of the thumb can be adjusted according to the object size. The thumb will rotate around the virtual centre of its MCP joint. From fully extended position to fully flexed position, the finger assembly provides 55° and 65° range of motion (ROM) for the MCP and PIP joints respectively, and the two joints will rotate together to help the hand perform close and open movements. The hand robot design can accommodate different palm size and finger length. It is a portable design where the total weight is 489 g (Ho et al., 2011). The finger positions and the palm position can be well fixed by belts with velcro on the hand robot. Users may carry it to practice hand functions with daily living activities (Ho et al., 2011). The hand opening motion is controlled by the EMG signal from the extensor digitorum (ED) muscle, which is the extensor for finger and palm extension; and the hand closing movement is controlled by the EMG from the abductor pollicis brevis (APB) muscle, which is a flat and thin muscle located just under the skin, whose activity is easy to be detected by surface EMG, and usually will contract during a hand grasp movement when holding an object (Moore and Dalley, 2006). In comparison with other fine muscles for finger movements, the EMG of APB and ED are less affected by spasticity (discussed later) and are relatively easier to be controlled by persons after stroke. A pair of EMG electrodes (Blue Sensor N, Ambu Inc. with a contact area of $20 \times 30$ mm) are on the skin surface of the muscle belly of a targeting muscle with a center distance of 2 cm. The reference electrode is attached to the skin surface of olecranon. Computer-based (Labview) control platform was developed with an EMG-triggered algorithm. It is based on real-time detection of the EMG levels of the respective muscles with a sampling frequency of 1000 Hz. Before the digitization, the EMG signals are first amplified with a gain of 1000 (amplifier: INA 333, Texas Instruments Inc.), then filtered with a band-pass filter from 10 to 500 Hz. After the digitization, the EMG signals of the APB and ED are rectified and low-pass filtered (10 Hz cutoff frequency with forth-order, zero-phase forward and reverse Butterworth filter) to obtain the envelope of the signal, i.e. the EMG level. The motor movement is initiated by the EMG levels of APB and ED for respective hand closing and opening, once the mean EMG level of a muscle with a length of 100 ms exceeded a threshold pre-set before each session; then, the motor of all the fingers will move at a constant angular velocity of 22°/s with the virtual center of the MCP joint and 26°/s with the virtual center of the PIP joint (The thumb only has the MCP joint). A common motor disability in stroke survivors is unable to voluntarily open the paretic hand, and the fingers are always in flexed positions. Therefore, the design of the robot hand is mainly for practicing the hand open and close functions on holding objects in this work.

### 2.2. Upper limb training with the EMG-driven robot hand

Ten subjects with chronic stroke were recruited after obtaining approval from the Human Subjects Ethics Sub-Committee of the Hong Kong Polytechnic University. The inclusion criteria of the study are: (1) the subjects were at least six months after the onset

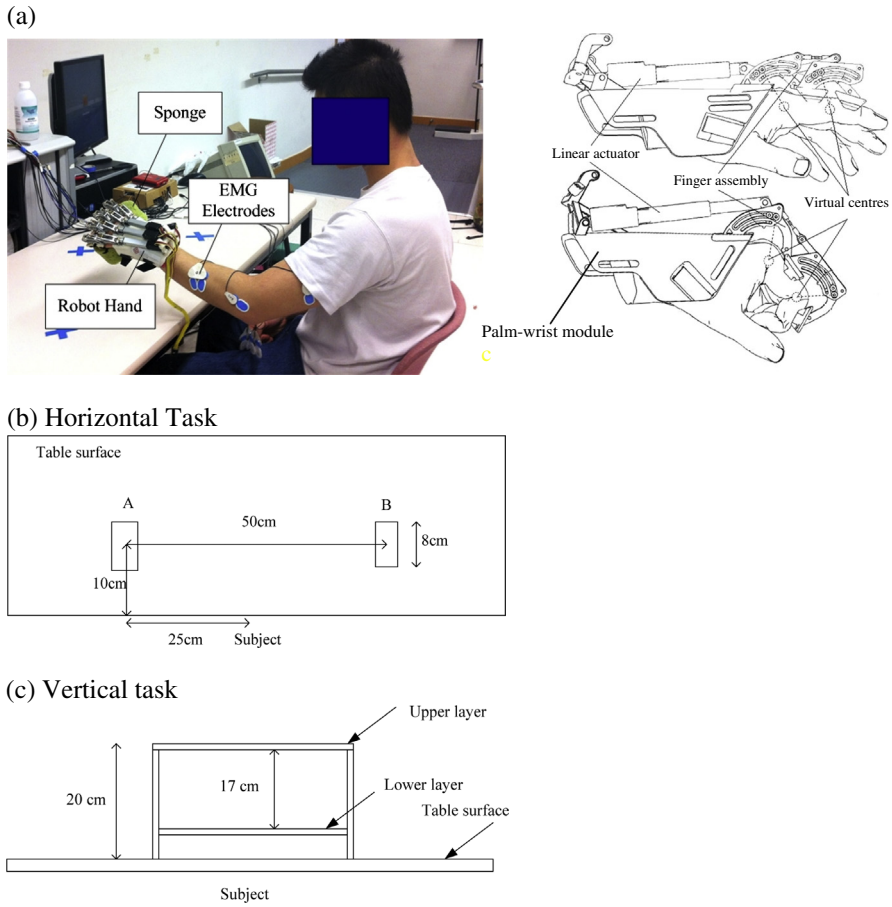


**Fig. 1.** The experimental setup. (a) The experimental setup for the EMG-driven robot hand assisted upper limb training and the schematic illustration on the mechanical structure of the robot hand; (b) the table configuration during horizontal task for the training; (c) the table configuration during vertical task for the training.

of a singular and unilateral brain lesion due to stroke, (2) both the MCP and PIP joints could be extended to 180° passively, (3) the spasticity during extension at the finger joints and the wrist joint was lower than 3 as measured by the Modified Ashworth Scale (MAS) (Ashworth, 1964), (4) there were detectable voluntary EMG signals (i.e. the signal amplitude should exceed 3 SD above the mean of the baseline) from the APB and ED in the paretic side of the subjects, and (5) the subjects also had sufficient cognition to follow the experimental instructions. The demographic data of the recruited subjects was summarized in Table 1.

Each subject was invited to attend a 20-session hand robot assisted upper limb training with a training intensity of 3–5 sessions/week and within 7 consecutive weeks. In each session, a subject first was required to conduct the MVCs at the muscles of APB, ED, flexor digitorum (FD), the biceps brahcii (BIC), and the triceps brachii (TRI) for two times. During the MVC test on the APB, FD and ED muscles, a subject was seated in front of a table by placing the paretic upper limb on the table with an extended elbow joint at 130° and with the wrist in the mid-position. The finger positions were held by an experimental operator with the MCP joints at 150° when conduct the MVCs. When doing MVC on the BIC and TRI muscles, the paretic upper limb was held by the operator with the shoulder abducted at 70° and the elbow flexed at 90°. Each MVC was maintained for 5 s, and 2 min rest was allowed between two consecutive contractions to avoid fatigue. In each training session, a subject was seated in front of a table with a position that the height from the surface of the table to the shoulder of the subject was from 30 cm to 40 cm. The subject was first required to conduct two bare hand evaluation tasks by using the paretic upper limb without wearing the robot system:

**Table 1**
Demographic data of the recruited subjects. Abbreviations: hemorrhage (H), infarct (I), left (L), right (R), female (F), and male (M).

| Subject no. | Nature of stroke | Affected UE | Gender | Dominant hand | Age (yrs) | Time since stroke (yrs) | Stroke severity (Fugl-Meyer upper limb, Max = 66) |
|---|---|---|---|---|---|---|---|
| 1 | H | L | F | R | 28 | 4 | 35 |
| 2 | H | R | M | R | 42 | 14 | 29 |
| 3 | I | L | F | R | 63 | 4 | 44 |
| 4 | H | R | M | R | 64 | 14 | 25 |
| 5 | H | L | M | R | 49 | 8 | 25 |
| 6 | I | L | M | R | 58 | 3 | 23 |
| 7 | I | R | M | R | 64 | 4 | 20 |
| 8 | I | L | M | R | 51 | 10 | 17 |
| 9 | I | L | M | R | 61 | 10 | 29 |
| 10 | I | L | F | R | 52 | 8 months | 33 |
| Mean ± SD | | | | | 53.2 ± 11.5 | 7.2 ± 4.7 | 28 ± 7.9 |

(1) Lateral arm reaching and grasp (horizontal task): For left hemiplegia, a subject grasped a sponge with a thickness of 5 cm and transported 50 cm horizontally from point A on the left of a table and to point B on the right of the table (Fig. 1b). The subject released the sponge at point B, and then grasped it again and returned to point A. For right hemiplegia, a subject completed the same sponge transporting cycle as for the left hemiplegia, but started the task from point B to point A. During the arm transportation, the testing hand was lifted up from the table with a distance (2–5 cm) away from the lowest point of the hand to the surface of the table.
(2) Vertical arm reaching and grasp (vertical task): A subject started the task from grasping the sponge on the mid-line of the lower layer of a shelf, lifted it 17 cm vertically, and put it on the upper layer, then picked it up and put it back to the lower layer (Fig. 1c). During the reaching, the elbow joint varied within the range of 40° of flexion to 180° of extension.

Each horizontal or vertical task was repeated for three times. The subjects were also instructed to use their natural speed to perform the evaluation tasks. The main objective of the robot hand assisted rehabilitation is that the subjects could perform limb tasks without the help of the device. The bare hand evaluation in each session could reveal the progress of the recovery of the upper limb motor function without external assistance across the training sessions. In an evaluation trial, the start of the EMG recording was when a subject received the command from the operator, and the end of the recording was when the sponge was released from the testing hand at the target position. All the subjects could grasp the sponge during the bare hand evaluation across the sessions. However some ($n$ = 5) could not release the sponge in the early sessions of the training, and they were instructed to try their best to perform the hand open. Therefore, we set a 10-s maximum time limit: If the paretic hand could not release in 10 s, then they were allowed to use the unaffected hand to take off the sponge. Only the EMG signals within the 10 s were included for analysis. At the 20th session of the training, two of the five could release the object by the paretic hand without help.

After the bare hand evaluation tasks, the subject was instructed to conduct the lateral/vertical arm reaching and hand close/open tasks respectively for 10 min with the EMG-driven hand robot on the paretic side with the same seating configuration as for the evaluation. In this work, the training intensity in each session was standardized by the training duration when the tasks were conducted with a natural speed of individual subjects. The mean repetition of the lateral task within the 10 min across the training sessions was 15.7 ± 7.8 times, and the mean repetition of the vertical task was 12.4 ± 5.5 times. A subject could have 20 min rest after the first 10 min training. Some of the subjects recruited in the study had difficulty in lifting up the affected upper limb when wearing the robot hand mainly due to the weakness at the shoulder and the elbow. In order to compensate the additional weight from the robot hand, the subjects were allowed to use the unaffected side to provide self-aware minimum support at the wrist of the affected side for arm transportation. The weight effect from the robot hand would be discussed later. Fig. 2 shows the representative raw EMG trials during a horizontal evaluation task, and the illustration of APB EMG trial during the robot assisted hand closing task in training session 1 from subject 2. In Fig. 2a, it could be observed that APB, FD and ED contracted together in the task (i.e., co-contraction), and BIC kept firing during almost the whole process (mainly due to the involuntary spasticity). The EMG threshold in each session for individual subjects was a value that a subject could successfully trigger the motor of the robot hand by their voluntary EMG signals. The threshold value would be gradually increased once a driving muscle (i.e., ED for hand open and APB for hand close) could generate stronger EMG signals. The minimum value of the threshold was 3 times of the standard deviation (SD) above the mean value of the EMG baseline during the resting state of the paretic hand (Ho et al., 2011), and the maximum value was 10% of the EMG level during MVC of a driving muscle throughout the training. Fig. 2b is an illustration on two APB EMG triggering cycles for hand close of the robot, when the threshold was set at 10% MVC. The 10% EMG level of MVC was an empirical maximum value that most of the subjects could finish the whole set of trials without feeling too difficult or too easy to achieve. In this work, the triggering thresholds of the driving muscles of ED and APB were also tested after wearing the robot hand in each session before the training in a testing trial for the respective horizontal and vertical tasks. All subjects could successfully trigger the hand close/open of the robot hand by their own muscular efforts from the APB and ED muscles in the testing trial, whereas no robot hand movement was triggered during the resting of the muscle. The period of a training session was within 1 h, including the bare hand evaluation.

### 2.3. Evaluation on the training effects

Before and after the hand robot training, clinical assessments were conducted by a blinded assessor. The assessments used in this study were Fugl-Meyer (Fugl-Meyer et al., 1975) (Fugl-Meyer with the full score of 66 for the upper limb assessment was further derived into shoulder/elbow (42/66) and wrist/hand (24/66) subscales as in (Takahashi et al., 2008; Krebs, 2013)), Action Research Arm Test (ARAT) (Carroll, 1965), MAS on the flexors related to the elbow, wrist, and fingers (Ashworth, 1964), and Wolf Motor Function Test (WMFT) (Wolf et al., 1989). Fugl-Meyer measures the motor functional impairment in voluntary limb movements after stroke, i.e., stroke severity. ARAT evaluates the upper limb functions with the hand tasks to hold and release objects with different shapes, sizes and weights. MAS measures the spasticity (or muscle tone) of a muscle. In this work, the muscle tone was only captured in the flexors of the recruited subjects, and the values of MAS presented are for flexors. WMFT provides information on the motion speed and functional ability of the upper limb when doing daily tasks.

In order to monitor the training progress across the sessions, EMG signals from the FD, BIC and TRI muscles were also captured during the bare hand evaluation, together with those from the APB and ED muscles in each session. Two EMG parameters were calculated for the cross-session monitoring, i.e., (1) EMG activation level of each muscle, and (2) co-contraction index (CI) between a muscle pair. These two parameters had been used for quantitative monitoring the muscle activation and coordination pattern changes in our previous studies (Hu et al., 2007, 2009, 2008), and the processing methods were as follows: All EMG signals were sampled with 1000 Hz, and stored for off-line process. The linear envelope of the recorded EMG signals was first calculated by: (i) full-wave rectification, (ii) low-pass filtering (10 Hz cutoff frequency with forth-order, zero-phase forward and reverse Butterworth filter), (iii) subtraction of the average EMG activity during the resting state, and (iv) normalized to the maximum value of EMG activation in each muscle during the MVC in each session. The EMG activation level of a muscle was calculated by averaging the EMG envelope of the trial (Hu et al., 2009, 2008), i.e.,

$$\overline{\mathrm{EMG}} = \frac{1}{T}\int_0^T \mathrm{EMG}_i(t)dt \tag{1}$$

where $\overline{\mathrm{EMG}}$ was the EMG activation level of muscle $i$, $\mathrm{EMG}_i(t)$ was the EMG envelope signal, and $T$ was the length of the signal.

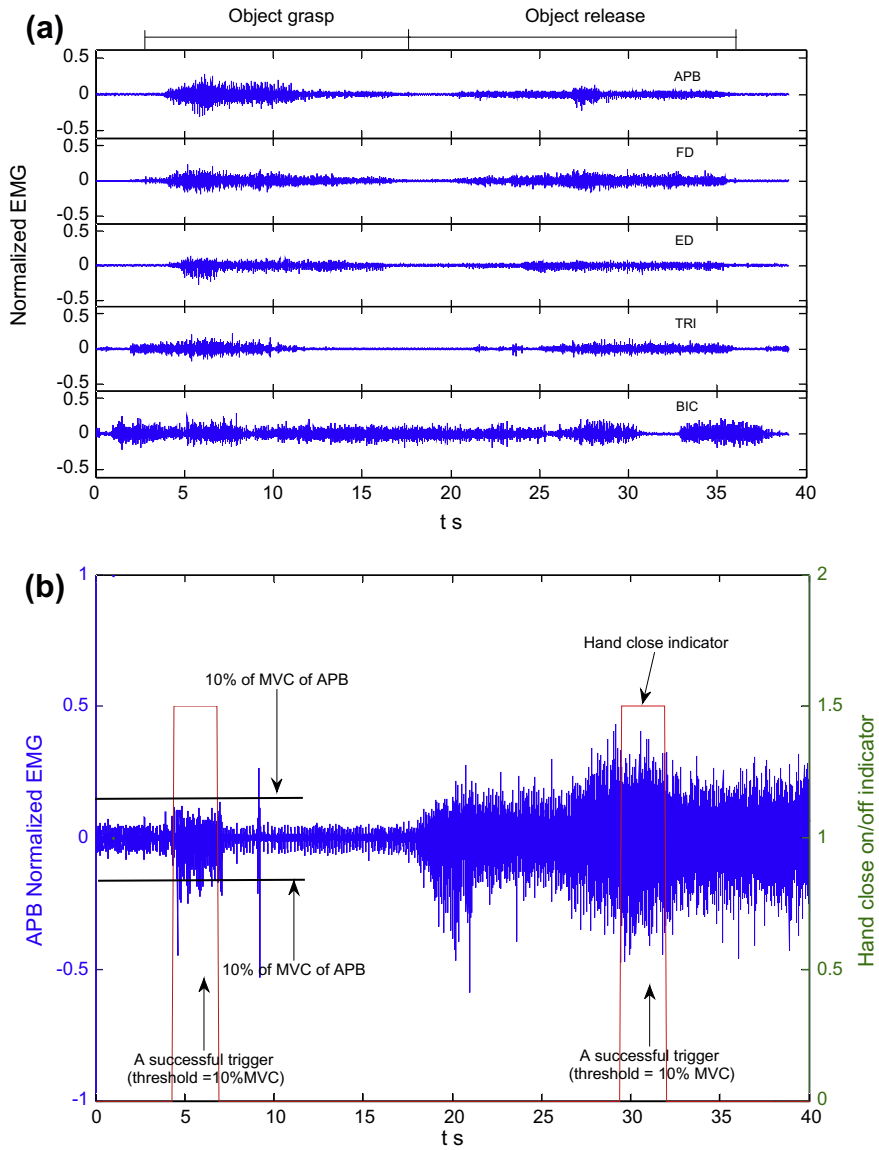


**Fig. 2.** Representative EMG trials of subject 2 in training session 1 during (a) the evaluation (a horizontal task trial), (b) the illustration on the EMG triggering threshold during hand close by APB EMG signal, when the threshold is 10%MVC.

The co-contraction index between a pair of muscle could be expressed as (Hu et al., 2009, 2008; Frost et al., 1997),

$$\mathrm{CI} = \frac{1}{T}\int_0^T A_{ij}(t)dt \tag{2}$$

where $A_{ij}(t)$ was the overlapping activity of EMG linear envelopes for muscles $i$ and $j$, and $T$ was the length of the signal. The value of a CI for a muscle pair varied from 0 (non-overlapping at all in the trial) to 1 (totally overlapping of the two muscles with both EMG levels kept at 1 during the trial). An increase of CI value suggested an enlarged co-contraction phase of a muscle pairs; while a decrease usually suggested a separation in the co-contracting phase of the two muscles. In a session, the evaluation task was repeated three times. It was expected the robot hand assisted training would modify the co-activating patterns of the muscle group in the whole upper limb. Muscle pairs of all possible combinations with the muscles of interest were studied by CI for understanding the co-activation pattern within the same joint and across the joints. Monitoring the varying patterns of the EMG parameters across the training sessions would provide a better understanding on the progress of the training effects on the upper limb functions.

### 2.4. Statistics

Normality test was performed on both the clinical scores and the EMG data by Lilliefors method with significant level of 0.05 (Mohd Razali and Wah, 2011). It showed that the EMG samples had normal distribution ($p < 0.05$, for the bare hand evaluation, during the horizontal and vertical training), but the normality of the clinical scores was not significant ($p > 0.05$, for the pre- and post- training evaluations). Therefore, Wilcoxon test was performed on the clinical scores, by a paired comparison on the data before and after the training for the same subjects. The one-way analysis of variance (ANOVA) with repeated measures (Bonferroni Post hoc test) were used to evaluate the training effect on the EMG levels and muscle co-contraction indexes across different sessions during the training. The overall EMG levels of each muscle (for the total 20 sessions) in the bare hand evaluation and the robot hand

assisted training were also analyzed by paired *t*-test. The level of statistical significance was set at 0.05 in this study.

## 3. Results

### 3.1. Clinical assessments

The clinical scores before and after the hand robot assisted upper limb training were compared in Fig. 3. Significant improvements were found in Fugl-Meyer shoulder/elbow ($P = 0.0078$), Fugl-Meyer wrist/hand ($P = 0.0039$), ARAT ($P < 0.001$), and WMFT ($P = 0.0156$). The MAS_finger was also significantly reduced after the training ($P = 0.031$). Decreases in the mean values of MAS_elbow and MAS_wrist were observed but without significant difference ($P = 0.084$ for MAS_wrist and $P = 0.109$ for MAS_elbow). From WMFT Time, it was also found that the time of performing evaluation tasks in the affected upper limb also significantly decreased after the training ($P = 0.007$). According to the data records of ARAT, the main improvement was on items of finger pinch on a marble or a ball (6 mm in the diameter). Before the training, six recruited subjects could not perform the pinch by the paretic hand in ARAT tasks. However, after the training all subjects could achieve pinch on the marble by using the thumb and the index finger, and six of them could conduct the pinch tasks by using the thumb and the middle fingers.

### 3.2. Cross-session EMG parameters

Fig. 4a shows the variations with statistical significance in EMG activation levels of the muscles during the bare hand evaluation tasks and across the training sessions. Significant decrease in EMG level was found in the ED and FD for the horizontal evaluation ($P < 0.05$, 1-way-ANOVA with post hoc tests). The varying trend of an EMG trace was determined based on the mean comparison by post hoc tests of ANOVA with significant level of 0.05 on a consistent increasing or decreasing tendency. An increased trend was determined by that 50% sessions from 6th to 20th had significant higher values than the maximal in the first five sessions (i.e., session 1–5), and a decrease was that 50% sessions from 6th to 20th had significant lower values compared with the maximal in the first five sessions. By comparing the mean values of the first and the last sessions (i.e., pre- and post comparison), the EMG level of ED dropped 31.1% and 17.2% for the FD muscle. Significant decrease in BIC EMG level and significant increase in TRI EMG level were observed during the vertical evaluation ($P < 0.05$, 1-way-ANOVA with post hoc tests). The TRI EMG level increased 36.1%, and the EMG level decrease in BIC was 12.1%. There was no significant decreasing or increasing trends observed in other cross-session EMG levels. Fig. 4b shows the variations with statistical significance in co-contraction indexes of the muscle pairs during the evaluation across the training sessions. The CIs for the muscle pairs of FD&ED, ED&BIC, FD&TRI, FD&BIC, and ED&TRI decreased significantly in the evaluation of horizontal task across the sessions ($P < 0.05$, 1-way-ANOVA with post hoc tests). By comparing the first and the last sessions, the decrease in the CI of FD&ED was 33.4%, ED&BIC was 23.6%, FD&TRI was 27.8%, ED&TRI was 41.8%, and FD&BIC was 28.9%. The co-contraction for the muscle pairs of FD&TRI, BIC&TRI, and APB&TRI increased significantly during the vertical task evaluation ($P < 0.05$, 1-way-ANOVA with post hoc tests). By comparing the mean values in the first and last sessions, the increase of CIs in FD&TRI was 95.8%, in BIC&TRI was 51.5%, and in APB&TRI was 31.6%. There was no significant increasing or decreasing trends observed in other cross-session CIs.

Fig. 5 shows the comparison on the overall EMG levels (for the total 20 sessions) of the muscles during the bare hand evaluation and during the robot hand assisted training. It was found that for the horizontal task, the EMG levels of APB, ED and FD during the robot hand assisted training were significantly high than those during the bare hand evaluation ($P < 0.05$). For the vertical task, the EMG levels of APB and ED were significantly higher than those during the bare hand evaluation ($P < 0.05$). However, there was no difference found in the other muscles.

## 4. Discussion

### 4.1. Training effects by clinical assessments

Most of the stroke subjects recruited in this study have spastic joints in the affected upper limb, especially at the fingers associated with an uncontrollable grasping hand before the training.

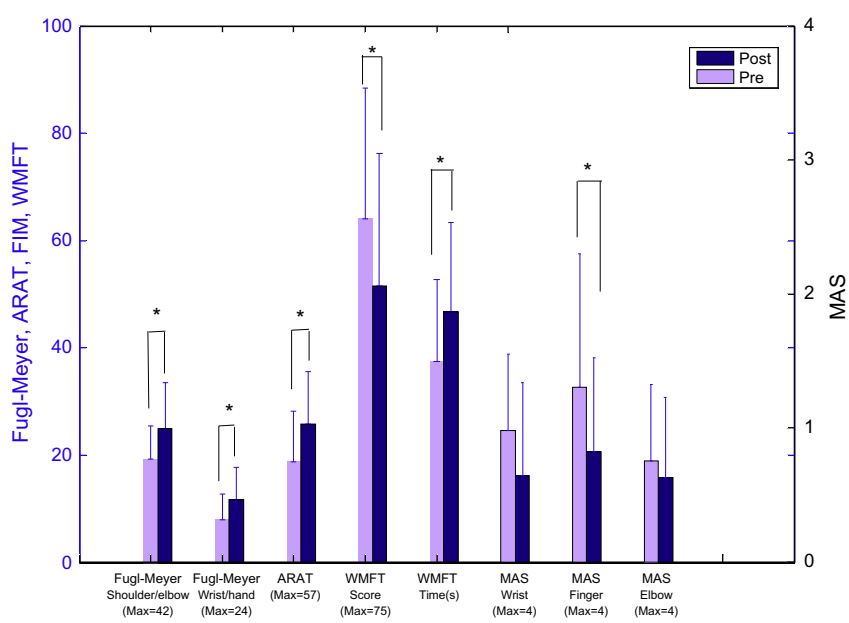


**Fig. 3.** Clinical scores pre- and post- the training represented by the means and standard deviations. The means with significant difference ($p < 0.05$) between the pre- and post- tests are marked with '*'. Significant improvements were observed in the Fugl-Meyer (shoulder/elbow, wrist/hand), ARAT, WMFT (time and score) and MAS_finger after the training.

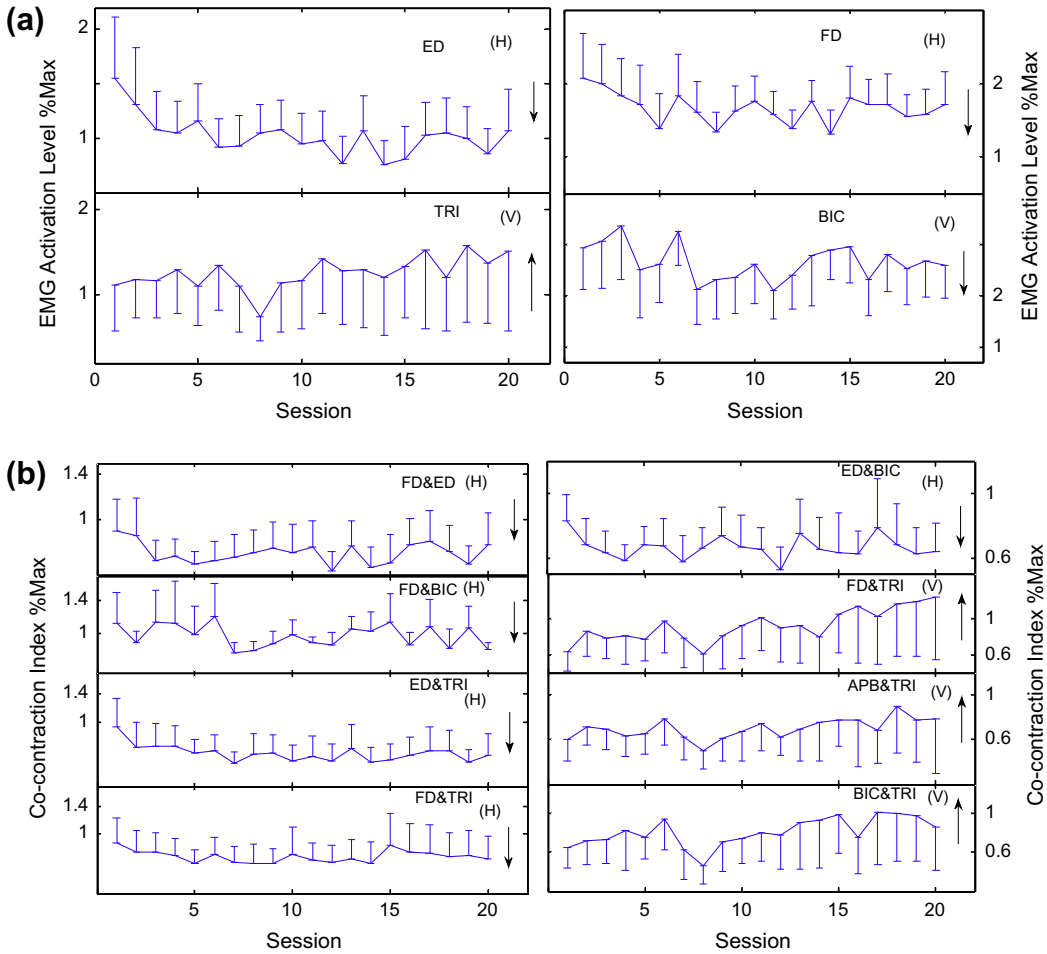


**Fig. 4.** (a) Varying patterns in the EMG levels with statistical significance ($P < 0.05$ with one-way-ANOVA with post hoc tests) during the bare hand evaluation. (b) Varying patterns in the CIs of the muscle pairs with statistical significance ($P < 0.05$ with one-way-ANOVA with post hoc tests) during the bare hand evaluation. (H) and (V) are for the horizontal and vertical evaluations respectively. The up and down arrows represent increasing and decreasing trends respectively.

They also demonstrated rigid muscle co-activation patterns when doing the bare hand evaluation and robot-assisted training in early sessions. The results of the clinical assessments suggested that the EMG-driven robot hand assisted training improved the voluntary motor functions in the affected upper limb. The increased Fugl-Meyer (shoulder/elbow, wrist/hand) suggested the training improved the voluntary motor control at the related joints, especially on the joint stability and range of motions of the shoulder and elbow, and also on the coordination of fingers during grasping. The ARAT assessment mainly evaluated the motor functions of the hand and the whole arm. Although previous studies have reported that the ARAT scores could have high correlation with the score of Fugl-Meyer for upper extremity assessments (Rabadi and Rabadi, 2006; Wei et al., 2011), ARAT test was more related to the hand function, 16 out of totally 19 items were related to the hand movements in the assessment. The increased score in ARAT on pinch tasks after the training also indicated the improved finger coordination. The increased scores in WMFT suggested the overall improvements in the whole upper limb, especially on the hand grasping tasks, which was consistent with the observations in the Fugl-Meyer and ARAT tests. The decreased time in conducting WMFT tasks (WMFT_Time) suggested that the movements of the paretic limb for achieving the required tasks were faster after the training with more effective muscle coordination. The reduction in the MAS_finger indicated a release of the flexor spasticity in the finger joints. It also suggested that by using the hand robot, it was effective to reduce the joint spasticity, which was consistent with other robot assisted trainings on different joints of the upper limb (Volpe et al., 2008, 2004; Takahashi et al., 2008; Lambercy et al., 2011). The task oriented whole upper limb training in this study was effective to improve the voluntary motor functions. The spasticity reduction at the elbow and the wrist was not that significant as at the fingers. In this study, there was no robotic support to the elbow and wrist joints to formulate the range of motions of the two joints during the training. It might imply that voluntary physical practice assisted by robot with well defined joint movement could be more effective in releasing spasticity and motor coordination.

### 4.2. Training effects by cross-session EMG monitoring

#### 4.2.1. Bare hand evaluation

The training tasks adopted in this work were grasping and releasing on an object horizontally and vertically, which could simulate the common scenarios in the daily living. The motor function restored after stroke actually is the motor skill acquired through repeated physical practice during the training. Therefore, the evaluation tasks in the work were same as the training tasks, but without the assistance from the robot hand. Individual muscle activation and coordination patterns among the firing muscles could be monitored by EMG signals. For the results of the bare hand evaluation, the possible reasons for the significant changes in the EMG levels of ED, FD, BIC, and TRI (Fig. 4a) are: (1) reduction of the spasticity in the flexors associated with a decrease in MAS scores after the

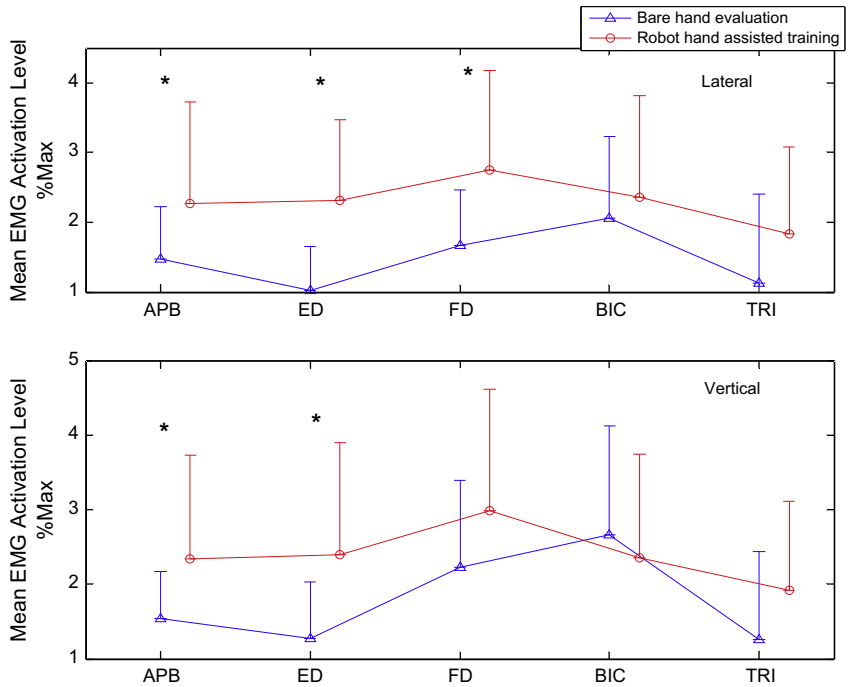


**Fig. 5.** Comparison on the overall EMG activation levels of the muscles of APB, ED, FD, BIC and TRI during robot hand assisted training (circles with solid lines) and during the bare hand evaluation (deltas with dotted lines). By paired *t*-test, significant differences ($P < 0.05$) on the mean EMG activation level are marked by '*' (with significant level set at 0.05).

training, and (2) a better muscular coordination for achieving the task with reduced CI values. The reduction in EMG level of the finger flexor (i.e., FD) also could be reflected by the reduced MAS_finger score after the training. It was observed that the reduction in FD muscle reached to a plateau around session 8, and that for the ED muscle was around session 12 during the horizontal task evaluation. The reduction of ED and FD resulted in the decrease of the CI of FD&ED (Fig. 4b), associated with the better performance of the hand open and close motions observed (i.e., opening the hand and releasing the sponge without help and improvements in the related ARAT, WMFT and Fugl-Meyer scores). The muscle coordination in upper limb were also improved during the training, as indicated by the decrease in the CI values of these muscle pairs (FD&TRI, FD&BIC, ED&TRI, and ED&BIC) during the bare hand horizontal evaluation (Fig. 4b). The value of CI was task dependent. For the extension and flexion of a joint, a reduction in the CI of the joint related antagonist muscle pair suggested a separation of the co-contraction phase, i.e., the muscle pair could contract more independently. For tasks, such as load bearing (e.g., against the gravity) or static position holding, CI of an antagonist muscle pair would increase to keep the related joint stable. The decrease of CI value of a muscle pair also suggested that the two muscles could contract more independently for more possible synergic patterns. Excessive co-contractions from the proximal joints when doing hand open and close were commonly observed in stroke survivors. Reduction in CIs in cross-joint muscle pairs suggested reduced compensation movement from the co-contraction on the proximal joints when doing hand open and close tasks. In this work, the decreases in CI value of the muscle pairs related to the FD and ED muscles also indicated a better motor control on the finger movements during the hand open and close, and more independent hand functions separated from the muscle co-contraction compensated from the proximal joint, e.g., the elbow. However, neither reduction in FD&ED co-contraction nor reduction in EMG levels of the FD and ED muscles was observed in the bare hand evaluation of vertical task. One of the reasons could be that more muscular efforts were needed for the fingers to hold the sponge still and to lift it against the gravity for the up and down movements. Spastic BIC and weak TRI were commonly observed in stroke survivors. During the vertical task, the subjects conducted the forward reaching and arm withdraw, which required more muscle effort for elbow extension and flexion by BIC and TRI muscles than in the horizontal task. The reduced BIC EMG level and increased TRI EMG level in the vertical task suggested the reduction in BIC activities (mainly due to the spasticity and excessive muscle efforts) and increased muscle power in the TRI during the arm reaching movements across the training sessions. However, it was found that the increase of the TRI EMG level did not reach a plateau within the 20 training sessions. It implied that more training sessions possibly may result in better motor improvement at the elbow joint. The increase of the TRI EMG level also was related to the increased CIs of APB&TRI, FD&TRI and BIC&TRI in the vertical evaluation tasks. From the results in the bare hand evaluation, it could be seen that the training would improve the coordination between the FD and ED during hand grasp and open (evaluated in the horizontal tasks), and also could result in a reduced BIC activity and an increase in TRI muscle power (evaluated in the vertical tasks).

#### *4.2.2. Robot hand assisted training*

It has been known that muscle spasticity and weakness are usually co-existed in the paretic limbs after stroke (Nakayama et al., 1994). Commonly, the flexors have more spasticity and the extensors have more weakness. However, the APB is a thumb flexor that had been found lack of spasticity after stroke (Towles et al., 2010). Therefore, in the design of the robot hand, the APB and ED were selected for controlling the open and close of the robot hand with less contamination of spastic EMG. It was also understood that the weight of the robot hand could affect the EMG signal during the training. The testing hand of a subject was usually in a stably pronated position when doing the training tasks, and the wrist joint was fixed by the palm–wrist module (Fig. 1). The weight of the robot hand mainly affected the EMG level of the muscles across the elbow and shoulder joints and these were related to lifting up the hand and forearm. The weight of the robot hand would not

affect the finger muscles (i.e., the ED, the APB and the FD in this study), since there was no additional gravity load from the robot hand on the finger joints according to the design of the mechanical linkage of the robotic fingers and their connections to the robotic palm (Fig. 1). It was understood that not all of stroke subjects could lift up the paretic upper limb to perform the training tasks as wearing the robot hand. Therefore, the subjects in this study were allowed to use their unaffected arm to give the minimum support to the affected limb at the wrist joint against the gravity during the robot hand assisted training. From the results of Fig. 5, it was found that the weight of the robot hand did not significantly affect the EMG levels of the BIC and TRI (related to the elbow joint), as the EMG level of the two muscles during the training were not significantly varied from those during the bare hand evaluation; While the EMG levels for finger muscles were significantly higher during the robot assisted training than during the evaluation. It suggested that the support from the unaffected arm to the paretic side minimized the weight effect of the robot hand on the EMG of the elbow muscles monitored in this work. The increased EMG levels of finger muscles after wearing the robot hand was because that the subjects needed to deliberately exert muscular effort of the finger muscles for obtaining the assistive motions from the robot hand, e.g., the triggering threshold of 10% MVC. It also implied that the upper limb training incorporated with the EMG-driven robot hand could encourage the subjects to generate more voluntary muscle effort for finger movements during the practice.

One of the limitations of the study is the small size of the subject group in the experiment. More subjects are recommended for future randomized clinical trials to compare the training effects between the EMG-driven robot hand assisted upper limb training and the conventional physical therapy, and to investigate normative values of the co-contraction indexes for stroke subjects with different levels of impairments in comparison with unimpaired groups. The training setup will be improved to reduce compensatory movements by belts to fix the body trunk when a subject is doing the tasks, which may reduce the variations in EMG signals. In the future robot hand assisted training after stroke, more alternative hand tasks can be adopted, such as pinch grasp or jar opening.

## 5. Conclusions

In this work, we investigated the effects and motor recovery during the EMG-driven robot hand assisted task-oriented upper limb rehabilitation training. The EMG analysis showed that the whole upper limb training incorporated with the EMG-driven robot hand could improve the muscle coordination between the antagonist finger muscle pair, i.e., the flexor digitorum and extensor digitorum, and also could reduce the excessive muscle activities in the biceps brachii. We also monitored the variations in the muscular coordination patterns by EMG parameters across the whole training sessions, which provided a quantitative way to understand the progress of the training. Significant reduction in the EMG levels was observed in the flexor digitorum and the biceps brachii which was related to a release of spasticity in the finger and the elbow joints.

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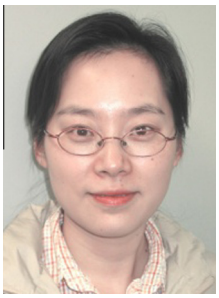

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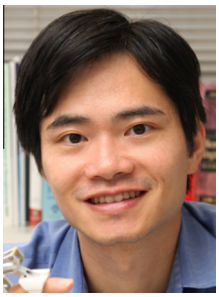

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